Beyond All-Equity Lifecycle Portfolios: A Public-Data Multi-Asset Extension

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Abstract

Should lifecycle investors hold only equities? Anarkulova, Cederburg, and O'Doherty (2025) find that internationally diversified stocks outperform conventional balanced and target-date allocations in their baseline model. We recover the domestic-to-international equity ranking in an annual public-data implementation, then widen the investment universe to currency-hedged global bonds, gold, and trend-following futures. Two simple four-sleeve families use fixed weights at exposures from 100% to 200%. Over 1927–2025, the equal-weight rule at 175% exposure reduces simulated retirement ruin from 6.98% to 2.79% and matches benchmark utility on 7.16% of income instead of 10%. At the same 175% exposure, the equity-only benchmark has 14.81% ruin and requires 9.64% saving. Under each of the three specified source influence diagnostics, this rule has lower annual volatility, lower ruin and a lower saving requirement than the benchmark. Both diversified families already reduce ruin without portfolio-level borrowing, although matching benchmark utility then requires more saving. Later historical windows also support the diversified ranking. Diversification and total exposure play distinct roles: combining asset classes improves capital preservation, while scaling the diversified portfolio lowers the saving required to finance retirement.

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1 Introduction

Households saving for retirement face two distinct decisions: which assets to combine and how much exposure to take. Balanced and target-date portfolios often change both at once. Adding bonds reduces the equity share and usually reduces total volatility, so the effects of composition and exposure arrive together. Lifecycle models place these choices in a household setting with labor income, longevity risk, retirement consumption, and bequests (e.g., Campbell and Viceira, 2002; Cocco, Gomes, and Maenhout, 2005). Anarkulova, Cederburg, and O'Doherty (2025), hereafter ACO, find that a fixed allocation of approximately 33% domestic and 67% international equities outperforms conventional balanced and target-date strategies in their baseline setting. Their opportunity set contains domestic and international stocks, local sovereign bonds, and local bills. Their leverage sensitivities also show that financing costs influence the optimal allocation.

How far does the equity result extend? ACO diversify stocks internationally but their baseline bond allocation holds local sovereign debt. We broaden the bond sleeve across issuers and add gold and trend-following futures. These assets bring different exposures to inflation, currency movements and market trends. Whether those differences improve retirement outcomes depends on their joint return distribution and on the cost of holding them.

The extension follows the distinction between portfolio composition and total exposure in Markowitz (1952) and Asness (1996). Diversification can improve the distribution of returns available to a household, while borrowing scales both gains and losses and incurs financing costs. We evaluate these choices together through two fixed portfolio families. Their annual volatility is a reported outcome of the comparison, alongside ruin and expected utility. We also scale ACO's equity allocation through the same exposure ladder, financing each borrowed dollar at the resident bill plus the same spread. This separates the benefit of broader diversification from the effect of borrowing more against equities.

Our annual replication uses the Jordà–Schularick–Taylor Macrohistory Database and supplementary public sources.[1] The final panel contains 1,561 country-years across 16 developed markets over 1927–2025. It recovers the qualitative ranking of the fixed equity and balanced strategies: simulated retirement ruin is 6.98% for the 33/67 benchmark, compared with 15.52% for domestic equities. The annual implementation provides a common framework for comparing these allocations with the multi-asset extensions.

The central experiment keeps the benchmark's equity mix fixed and adds the three other sleeves in two families: one proportional to 60/40/25/25 and one equally weighted. Each family is evaluated at gross exposures of 100, 125, 150, 175, and 200%. The portfolio weights follow simple rules specified independently of estimated means, covariances, and simulated utility.

The full-panel equal-weight allocation at 175% exposure reduces simulated ruin from 6.98% to 2.79% and equivalent saving from 10% to 7.16%. Under all three specified source-influence diagnostics, it has lower annual volatility as well as lower ruin and equivalent saving. Removing

[1] The research repository is available at `https://github.com/GF-boop/beyond-stocks`. Numerical outputs, replication scripts, and data provenance are detailed in Appendix G.

Italy 1942 gives 17.30% volatility against 18.17%, 2.61% ruin against 7.19%, and 7.02% equivalent saving against 10%. Removing France 1946 as well, or Italy throughout the sample, gives the same direction on all three measures. The diversified advantage therefore extends to volatility when the most influential source observations are removed.

The primary sample covers 1927–2025, retaining the range of monetary and market regimes in the historical record. Post-1950 and post-1970 windows examine the contribution of early history and preserve the favorable ruin and utility ranking of the two 200% families. Together, these comparisons show how a broader asset universe changes the trade-off between capital preservation and retirement resources.

2 ACO and the Multi-Asset Extension

2.1 The benchmark result

ACO evaluate retirement strategies in a lifecycle model with stochastic labor income, Social Security benefits, mortality risk, and a bequest motive. A couple works for 40 years and then draws down its retirement account. Their monthly return panel covers 39 developed countries over 1890–2023, with coverage varying by country. Multi-year bootstrap blocks can originate in different countries, while household earnings, demographics, and preferences remain anchored to a U.S. baseline.

Their baseline fixed allocation holds approximately 33% domestic and 67% international equities without borrowing. ACO also relax the leverage constraint: the optimal exposure and bond allocation depend on the borrowing spread. Our extension examines additional asset classes under explicit financing assumptions, whose sensitivity is reported in Section 5.

ACO select portfolio weights by maximizing average simulated utility. On our public panel, a grid search restricted to the domestic–international equity mix places the optimum at 25% domestic and 75% international, with 33/67 within 0.02 percentage points of equivalent saving. We therefore retain 33/67 as a common equity sleeve and reference strategy. The multi-asset families use fixed heuristic weights rather than an analogous optimization. The resulting comparison asks whether simple diversified rules improve on the equity allocation selected in this framework.

Expected utility already trades off favorable outcomes against adverse ones under the household's preferences. Comparing portfolios at similar annual volatility adds a complementary diagnostic: it helps distinguish changes in composition from changes in overall exposure. We use the fixed ladders of Section 5 for that purpose. The ladders report annual volatility alongside the lifecycle outcomes at each exposure level. Appendix E separately examines how the comparison changes with risk aversion.

2.2 Three empirical questions

Our investigation addresses three questions.

First, how much of the international-equity comparison survives when real exchange rates are held constant? We keep the national markets and weights fixed and remove the joint movement of nominal currencies and relative prices. The counterfactual measures the accounting contribution of currency movements.

Second, does replacing local sovereign bonds with a currency-hedged global bond basket improve outcomes? Keeping equity weights and leverage fixed allows us to compare the two bond sleeves under the stated currency and financing conventions.

Third, do the four-sleeve portfolios improve retirement outcomes across fixed exposure levels? We compare ruin, equivalent saving and annual volatility, then examine which conclusions survive changes in the historical sample, household preferences and implementation costs. This last step matters because extreme currency conversions can move the volatility comparison without reversing the lifecycle ranking.

3 Data

3.1 Country-year panel and resident numeraire

All empirical inputs to our lifecycle simulations are publicly accessible. Whereas ACO evaluate their model on proprietary monthly return series, we reconstruct the long-run developed-market record from open sources, ensuring that the replication and every extension can be reproduced from public data alone. The only proprietary series in the paper are the SG and BarclayHedge managed-futures indexes, which appear solely in the external validation of Appendix B.2 and enter no lifecycle simulation.

The panel begins with release 6 of the Jordà–Schularick–Taylor Macrohistory Database (Jordà et al., 2023; Jordà et al., 2019), which supplies nominal total returns on equities and long-term government bonds, short rates, inflation, exchange rates, and real GDP for 16 developed markets through 2020. We extend each series through 2025 using dividend-adjusted local indices, locally listed ETFs, and the Global Macro Database (Müller et al., 2025). Over the 2006–2020 overlap, the reconstructed equity series average approximately 1.2 percentage points below the Macrohistory series, primarily reflecting ETF management fees and large-cap benchmark tilts. A source indicator flags the appended years, allowing robustness checks to isolate the original Macrohistory window. The resulting baseline dataset is an unbalanced panel of 1,561 country-years.

An ex ante investability screen is retained as a labeled sensitivity. It excludes calendar years overlapping the prolonged closures or restrictions reported for our countries in ACO's Table A.III and years with CPI inflation of at least 50% or deflation of at least 20% in absolute value. These criteria identify observations for which an annual vector cannot synchronize a tradable market, a convertible currency, and a renewable hedge. None of them depends on portfolio returns. The replication package preserves the raw rows. Every calendar year from 1927 through 2025 remains represented.

Let $S_{c,f,t}$ denote units of resident currency c per unit of foreign currency f. If $R^N_{f,t}$ is a nominal foreign return and $\pi_{c,t}$ is resident inflation, the resident-real return is

$$1 + R_{f,t}^{(c)} = (1 + R_{f,t}^{N}) \frac{S_{c,f,t}}{S_{c,f,t-1}} \frac{1}{1 + \pi_{c,t}}. \tag{1}$$

We apply this spot conversion to unhedged assets before aggregating returns. Domestic equities, international equities, gold, and unhedged controls therefore all measure changes in purchasing power within the resident household's currency.

International equity excludes the country of residence and uses lagged market capitalization weights constructed from the Big Bang Database's historical market-capitalization-to-GDP ratios when coverage is sufficient (Kuvshinov and Zimmermann, 2022), with real-GDP weights (Maddison Project via the Macrohistory Database, extended by Müller et al. 2025) as a fallback. The 33/67 benchmark combines the resident's domestic market with this foreign basket. The constant-real-exchange-rate counterfactual instead aggregates each foreign market's local real return with identical weights. This accounting counterfactual excludes currency carry and hedging costs; it isolates the contribution of real exchange-rate movements.

The world bond block equally weights every sovereign in the panel with a usable bond, bill, and exchange rate in a given year. In 1,344 of 1,561 country-years it holds all 16 issuers; the floor anywhere is 13. Years with fewer than eight available issuers are dropped, which costs no observation after 1927. Like an actual global bond index, the block includes the resident's own sovereign. The basket is therefore common to all residents in a year, and each foreign bond uses the initial-notional forward of equation (4). Financing stays local throughout. We subtract ten basis points annually from covered notionals in the baseline. The unhedged foreign bond and bill basket is retained only as a labeled diagnostic control. The resulting basket spans the developed sovereign markets in the panel.

3.2 Gold and managed futures

Gold uses the annual U.S.-dollar price compiled by MeasuringWorth (Officer and Williamson, 2024) through 1967 and the LBMA afternoon fixing (London Bullion Market Association, 2025) from 1968, the first year a free market existed. Both are dollars per troy ounce, spliced without recalibration. We subtract 40 basis points per year for custody, convert the dollar return into each resident currency, and deflate by resident inflation. Before 1968, the dollar price was largely administered, and private U.S. ownership and convertibility were restricted. Under the gold standard and Bretton Woods, exchange-rate parities also constrained gold prices in other panel currencies. The early low-variance segment therefore reflects a monetary regime distinct from the subsequent free market (Erb and Harvey, 2013). The post-1970 window examines the portfolio comparison after this transition.

No free investable managed-futures index spans the sample. We reconstruct a monthly time-series-momentum proxy instead, following a well-established literature. Moskowitz, Ooi, and Pedersen (2012) document that the sign of an asset's own past twelve-month return predicts its next-month return across roughly 58 futures markets, and that scaling each position by inverse volatility and averaging produces a diversified premium. Hurst, Ooi, and Pedersen (2017) extend the same rule back to 1880 and show that a blend of one-, three-, and twelve-month look-backs is stable across a century of data and across asset classes. Nick Baltas and Kosowski (2020) turn to the construction choices themselves. Signal horizon, rebalancing frequency, and the resulting turnover jointly determine net performance and capacity. Hence our one-, six-, and twelve-month blend, and hence the explicit turnover cost below. Lempérière et al. (2014) find a trend premium in equities, bonds, currencies, and commodities as far back as 1800. That history is what justifies running the rule over the full 1927–2025 window. Trend following also lowers sequence risk in retirement decumulation (Clare et al., 2017; Clare et al., 2019) and improves risk-parity implementations through correlation-aware weighting (Nikitas Baltas, 2015). That literature, however, stops at portfolio-level backtests: none of it embeds trend following in a full lifecycle model with the pooled developed-market panel, which is the gap the lifecycle results of Section 5 fill.

The managed-futures signal uses a separate, frozen monthly price snapshot. Completed calendar-year returns are compounded and joined to the annual JST/Macrohistory household panel. Appendix B.1 documents that snapshot, its sources, and the monthly-to-annual transformation in detail.

In our implementation, signals average the signs of prior one-, six-, and twelve-month returns. Volatility comes from the preceding 36 months. The aggregate strategy targets 10% volatility under a three-times scalar cap. The universe expands across four sectors as data become available: 18 national price indexes and 18 synthetic ten-year sovereign positions, 19 commodities plus gold and silver, and six dollar foreign-exchange forwards. The monthly equity and yield inputs come principally from the OECD Main Economic Indicators via FRED (Organisation for Economic Co-operation and Development, 2026; Federal Reserve Bank of St. Louis, 2026). The early U.S., U.K., German, French, commodity, and rate segments come from the monthly NBER Macrohistory files (National Bureau of Economic Research, 2016), commodities are extended with the World Bank "Pink Sheet" (World Bank, 2025), and currencies use month-end Federal Reserve H.10 quotations (Board of Governors of the Federal Reserve System, 2026). Where two sources meet, the return that straddles the junction is left missing. This prevents source-level differences from entering the trading signal. Sector Sharpe ratios over the full reconstruction are 0.44 (equities), 0.50 (rates), 0.47 (currencies), and 0.36 (commodities), against 0.41, 0.49, 0.57, and 0.80 in the Capital Fund Management estimates of Lempérière et al. (2014). Combined, the figures are 0.77 against 0.80.

The proxy includes the U.S.-dollar collateral return recorded in the snapshot. The baseline treats its dollar NAV as a foreign asset and applies the initial-notional forward in equation (4), using the recorded collateral return for the foreign cash leg. Currency exposure on subsequent NAV gains or losses remains. This is the same hedge convention as for global bonds. Replacing collateral for the entire terminal value is the alternative full-value convention. A locally collateralized futures strategy would be a different implementation. We subtract a 0.85% annual management fee and transaction costs based on measured turnover of 18.88 times per year and a three-basis-point spread, or 0.57%

Table 1: Realized resident-real portfolio properties

Portfolio	Mean return	Volatility	1% annual return
Domestic equity	7.60%	23.34%	-45.64%
ACO 33/67	8.12%	25.28%	-40.36%
Stocks/I 50/50	7.99%	22.59%	-40.22%
Balanced domestic	5.38%	15.82%	-34.19%
Balanced/I	5.61%	15.04%	-27.63%
90/60 local	8.25%	23.95%	-36.55%
90/60 global bonds	8.19%	24.13%	-36.19%
ACO, constant real FX	7.44%	15.75%	-37.40%
60/40 ACO/covered	5.64%	16.51%	-25.67%
60/40 ACO + 33.33 MF	7.60%	17.90%	-21.66%
60/40 ACO + 33.33 gold	6.79%	23.86%	-23.33%
54/36/20/20 ACO	7.00%	19.87%	-19.67%
90/60/25/25 ACO	10.52%	30.27%	-29.94%

Note: Arithmetic means, standard deviations and linearly interpolated 1% quantiles of pooled annual returns across 1,561 country-years. Tabulated from the baseline simulation outputs. Bonds and managed futures use the fixed-notional hedge; managed futures include baseline fees and costs.

annually. Its monthly correlation with several official CTA indexes is 0.49–0.53 over 2000–2025.

3.3 Realized portfolio properties

Table 1 reports pooled one-year return moments from the same panel and hedge convention as the lifecycle results. The table reports mean returns, volatility, and the lower annual tail. Global bonds have slightly higher volatility than local bonds in the 90/60 comparison under this convention.

4 Lifecycle Model and Portfolio Design

4.1 Household, income, and retirement

The household is a couple aged 25. Each spouse receives an independent stochastic earnings history through age 64 based on model 6 of Guvenen et al. (2021). Log income contains a quadratic age profile, a persistent component with autoregressive coefficient 0.959, transitory shocks, and age- and income-dependent nonemployment. The household saves a constant fraction of each spouse's income when that individual's annual income is at least $15,000. Contributions arrive at the start of the year, in time to earn that year's portfolio return.

At age 65 the couple fixes a real annual withdrawal equal to 4% of retirement wealth, following Bengen (1994) and ACO. A fixed real withdrawal makes the retirement outcome sensitive to the order in which returns arrive, not only to their average: a sequence of early losses can exhaust the account even when long-run returns are adequate, and international evidence shows the safe rate varies widely across historical return paths (Estrada, 2018). Appendix F varies the withdrawal rate from 3% to 5% and the contribution rate from 5% to 15% and finds the diversified ranking unchanged. Social Security follows from each simulated earnings history, computed with the 2022 \$147,000 taxable-earnings ceiling, 2022 bend points, spouse and survivor benefits, and the early-claiming adjustments ACO describe. Supplemental Security Income supplies a consumption floor. Ruin occurs when financial wealth cannot fund the fixed withdrawal before the last spouse dies. Public pension income continues after financial wealth is exhausted.

Mortality follows sex-specific Gompertz–Makeham laws calibrated to the joint retirement and longevity moments reported by ACO from Social Security Administration tables (Social Security Administration, 2020). Return paths, income histories, and death draws are paired across strategies.

Following ACO, the model combines U.S. earnings, public pensions, and preferences with pooled developed-market financial history. Each simulated lifetime draws successive blocks from this history, exposing the same household to a wider range of economic environments than a single national record provides. Countries share shocks and global asset baskets, so the 1,561 country-years form a dependent panel. Every strategy faces the same return, income, and mortality draws within an experiment. Section 4.4 describes the resampling procedure.

4.2 Utility and equivalent saving

For annual real consumption C_t, household size H_t, bequest B, and last year alive T, utility is

$$U(C,B) = \sum_{t=65}^{T} \frac{(C_t/\sqrt{H_t})^{1-\gamma}}{1-\gamma} + \theta \frac{(B+k)^{1-\gamma}}{1-\gamma}. \tag{2}$$

Risk aversion is $\gamma = 3.84$, the bequest motive has intensity $\theta = 2{,}360$, and the shifter is $k = \$490{,}000$. These are the estimates of De Nardi, French, and Jones (2010), fitted to retiree data in a lifecycle model with medical expenses, and ACO adopt them unchanged. Our utility aggregator therefore uses their stated preference parameters, while the wealth and consumption dynamics remain an annual approximation of their monthly engine. The bequest term is additive and warm-glow: utility depends on the size of the estate left, $B + k$, not on the heirs' own circumstances, which is the specification that explains why retirees in the data run down wealth slowly. ACO evaluate monthly consumption and therefore multiply the annual bequest-intensity estimate by 12^{γ}. If within-year consumption is uniform, dividing their entire monthly utility by 12^{γ} gives equation (2) with $\theta = 2{,}360$. Under uniform within-year consumption, the annual aggregation thus preserves the relative scale of consumption and bequest utility.

For strategy j, the equivalent saving rate s_j^* solves

$$E[U_j \mid s_j^*] = E[U_{33/67} \mid s = 0.10]. \tag{3}$$

A rate below 10% means the strategy reaches benchmark utility on less saving. Every other reported outcome, wealth, consumption, ruin, and bequest, uses the common 10% rate and not s_j^*. The metric evaluates composition and exposure jointly. A lower-exposure portfolio can require more saving despite reducing the probability of ruin. The fixed-exposure ladders in Section 5 and

Table 2: Fixed portfolio rules

Portfolio	Equity	Bonds	Gold	MF	Sleeve sum	Financing
ACO 33/67	100%[a]	0%	0%	0%	100%	0%
90/60 local	90[a]	60[b]	0	0	150	50
90/60 global bonds, covered	90[a]	60[c]	0	0	150	50
60/40 ACO/covered	60[a]	40[c]	0	0	100	0
60/40 ACO + 33.33 MF	60[a]	40[c]	0	33.33	133.33	33.33
60/40 ACO + 33.33 gold	60[a]	40[c]	33.33	0	133.33	33.33
54/36/20/20 ACO	54[a]	36[c]	20	20	130	30
90/60/25/25 ACO	90[a]	60[c]	25	25	200	100

Note: Percentages of investor capital. [a]33% domestic and 67% international equities. [b]Residence-country government bond and bill. [c]Currency-hedged equal-weight global sovereign bonds with resident-currency financing. The equity sleeve remains 33/67. "Sleeve sum" is the sum of index-level allocations, and financing is the corresponding external cash financing in the return equation.

Appendix C therefore report utility alongside annual volatility. The joint comparison identifies what additional exposure contributes to retirement outcomes.

Higher risk aversion penalizes low household consumption rather than annual return variance itself. Social Security and the consumption floor moderate the transmission of portfolio losses to consumption, while large positive returns can raise measured volatility without creating low-consumption states. Equivalent saving also compares each rule with a benchmark whose utility is recomputed at the same γ. With annual bequest intensity held fixed, raising γ from 2 to 10 increases required saving from 3.86% to 5.38% for the equal-weight 200% rule, but from 9.23% to 17.60% for ACO at the same exposure. The response therefore depends on the consumption distribution, not on leverage alone (Appendix E).

4.3 Fixed, non-estimated portfolio rules

Table 2 defines the main extensions. The 90/60 local strategy holds 90% in the ACO 33/67 equity portfolio and 60% in the residence-country sovereign bond, financed with a 50% short local-bill position plus the financing spread ϕ on the borrowed portion. The intermediate strategy changes only the bond sleeve to the covered global aggregate, with financing still in the resident bill. Every other extension also uses the same 33/67 resident equity return. The designs therefore vary fixed income, alternative sleeves, and total exposure without changing equity geography.

The weights are design inputs fixed prior to evaluation.

The manifest also contains two fixed leverage ladders, each stepping a four-sleeve family through 100, 125, 150, 175, and 200% gross exposure: one proportional to 60/40/25/25 (equity/covered bonds/gold/managed futures), the other equal-weight across the four sleeves. The levels are fixed before evaluation and target no realized volatility. These ladders carry the central comparison in Section 5.

Let A_t^f be the foreign asset's gross nominal return, B_t^f and B_t^c the foreign and resident gross nominal bill returns, and $q_t = S_{c,f,t}/S_{c,f,t-1}$ the spot currency movement. The baseline covers the initial foreign principal with a one-period forward. Under the assumed covered-interest-parity forward ratio, its resident-real return is

$$1 + R_t^{c,H} = \frac{B_t^c/B_t^f + (A_t^f - 1)q_t}{1 + \pi_{c,t}}. \tag{4}$$

The asset gain or loss beyond the initial principal retains currency exposure. Annual realized bill returns proxy the interest rates used to price the forward. The former full-value hedge is retained only as a sensitivity. Let every $R_{i,t}^{(c)}$ then be in the common resident-real numeraire. A portfolio whose index-level sleeve sum is G_j has return

$$R_{j,t}^{(c)} = \sum_i w_i R_{i,t}^{(c)} + (1 - G_j) R_{f,t}^{(c)} - \max(G_j - 1, 0)\phi - h_j \kappa, \tag{5}$$

where ϕ is the financing spread, h_j is covered notional, and κ is the hedge friction. The managed-futures NAV includes its recorded U.S. collateral once; the forward payoff is added under equation (4). The portfolio financing position in equation (5) is separate. For ACO at exposure G, the same equation becomes $R_{\mathrm{ACO},G} = GR_{33/67} + (1 - G)R_f - (G - 1)\phi$ for $G \geq 1$. Its equity sleeve remains unhedged. We evaluate $G = 1, 1.25, 1.5, 1.75, 2$; the utility reference remains unlevered ACO at a 10% saving rate.

For a levered portfolio ($G_j > 1$), the coefficient on the resident bill in equation (5) is negative. Holding all asset returns fixed, a higher ex-post real bill therefore lowers the portfolio return, while an inflation surprise that lowers the ex-post real bill raises it. This effect follows directly from the accounting identity of annual rebalancing, abstracting from intra-year borrowing constraints.

The baseline financing and hedge charges are 30 and 10 basis points, respectively. Both enter as annual real-return deductions. Cost sensitivities vary these deductions while retaining the observed real bill return.

Gross exposure sums allocations to return sleeves; managed futures also uses variable derivative positions within its NAV. The annual model captures portfolio financing and hedge costs through equation (5).

Return-stacked ETFs now give individual investors access to combined exposures through ordinary fund shares. For example, the Return Stacked Global Stocks & Bonds ETF targets approximately one dollar of global equity exposure and one dollar of U.S. Treasury exposure per dollar invested (Tidal Trust II, 2026). Funds combining bonds or equities with managed futures provide further building blocks (Tidal Trust II, 2024). These structures bring portfolios of the kind studied here within reach without requiring the household to manage derivative positions or borrowing directly. When shares are purchased outright, without borrowing or pledging them as collateral, the investor faces no personal margin calls arising from the fund's leverage. The fund manages collateral and financing internally; the associated costs and any deleveraging affect its net asset value. Actual funds differ in their asset coverage and trading rules from the synthetic sleeves used here.

A complementary diagnostic examines direct borrowing in a personal margin account. With annual rebalancing and a 25% maintenance requirement, none of the four tested allocations breaches the threshold at a year-end across 10,000 simulated 86-year return paths, or 860,000 account-years per allocation. The minimum equity-to-assets ratio is 47.69% for the equal-weight 175% portfolio and 37.59% and 38.98% for the proportional and equal-weight 200% portfolios. Appendix H reports the calculation, including the 150% stock–bond comparator. This is a year-end solvency diagnostic; margin breaches within the year require higher-frequency data. The same test records 1,081 threshold breaches for ACO at 175% and 17,934 at 200%, corresponding to 0.13% and 2.09% of simulated account-years. These are threshold diagnostics under annual exposure resets, without modeling forced liquidation after a breach.

4.4 Bootstrap and inference

The main fixed-portfolio experiment draws 10,000 paths using a stationary block bootstrap with mean block length ten years, following ACO. Ten-year blocks preserve two things an i.i.d. resampling would distort: the horizon-dependent structure of equity returns, including the long-run mean reversion documented by Poterba and Summers (1988), and the persistence of real interest rates. The block advances through consecutive years of a country's history, preserving the joint asset-return vector. A completed block starts again at a uniformly sampled panel row. At a gap or series end, the unfinished block continues at the first available year of another country. The implementation selects that country through a uniformly sampled row, so countries are weighted by their available sample lengths. This implements ACO's continuation principle at annual frequency.

Our resident-numeraire conversion operates inside each block. Every asset on a country-year row already sits in that country's currency and price index (Section 3). A block drawn from, say, Japan in the 1980s is therefore a fully consistent yen-real decade: domestic and foreign equity, the hedged bond and managed-futures blocks, and inflation all measured in one household's purchasing power. The numeraire changes only when the next block comes from another country.

We report normal 95% intervals for the paired difference in ruin relative to the 33/67 benchmark. These intervals measure Monte Carlo uncertainty conditional on the observed panel. Appendix D separately reports the range of outcomes across historical windows and source-country omissions.

5 Results

5.1 Replication and the central comparison

The annual panel contains 1,561 country-years. We use the stationary bootstrap with the continuation rule described by Anarkulova, Cederburg, and O'Doherty (2025), retain historical observations except for documented market-closure and monetary-reform exclusions, and value the currency hedge as a one-period, fixed-notional forward. The baseline applies no return-dependent screen.

Table 3 gives the primary results. The public-data replication preserves the economic direction of the international-equity comparison: ACO 33/67 has lower simulated ruin than domestic equity.

Table 3: Baseline results

Portfolio	Annual volatility	Equivalent saving	Retirement ruin
Domestic equity	23.34%	18.60%	15.52%
ACO 33/67	25.28%	10.00%	6.98%
ACO 33/67, 175%	43.99%	9.64%	14.81%
ACO 33/67, 200%	50.28%	15.66%	20.28%
Stocks/I 50/50	22.59%	10.56%	7.78%
Proportional four-sleeve, 175%	26.47%	6.26%	2.25%
Equal-weight four-sleeve, 175%	25.31%	7.16%	2.79%

Note: 10,000 paired lifecycle simulations. Equivalent saving is the contribution rate that matches expected utility of ACO 33/67 at a 10% saving rate. The four sleeves are ACO equity, fixed-notional currency-hedged global bonds, managed futures, and gold. Both families use the same 175% gross exposure.

The equal-weight four-sleeve rule at 175% exposure reduces simulated ruin from 6.98% to 2.79% and utility-equivalent saving from 10% to 7.16%. The paired 95% Monte Carlo interval for the ruin difference is $[-4.67, -3.71]$ percentage points. Annual volatility is 25.31%, compared with 25.28% for ACO; the source-influence diagnostics below examine the concentration of these full-panel estimates.

5.2 Diversification and equity leverage

Increasing ACO's equity exposure initially lowers its saving requirement: the 125% and 150% rules require 8.56% and 8.30%, respectively. Beyond 150%, the saving requirement rises to 9.64% at 175% and 15.66% at 200%. Ruin increases throughout the ladder, from 6.98% at 100% to 20.28% at 200%.

At the same 175% gross exposure, the proportional and equal-weight portfolios have ruin of 2.25% and 2.79%, against 14.81% for levered ACO, and require 6.26% and 7.16% saving, against 9.64%. At 200%, their saving requirements fall further to 5.09% and 5.92%, while ACO's rises. Thus the gains cannot be reproduced simply by applying the same leverage to the equity benchmark. Equal gross exposure equates portfolio borrowing; annual volatility remains an outcome of composition. Appendix I reports the complete ACO ladder and the matched-exposure construction, cost, and household-policy sensitivities.

5.3 Which sleeves account for the result?

We examine whether the advantage over ACO extends to simpler compositions by removing global bonds, gold, and managed futures one at a time from each 175% family. The removed notional is reallocated pro rata among the three remaining sleeves, leaving gross exposure, borrowing, hedge convention, and per-unit cost assumptions unchanged. Total sleeve and hedge charges adjust with the resulting weights. The comparison measures the effect of changing composition at fixed exposure.

Table 4: Constant-exposure sleeve ablations at 175% gross exposure

Family	Composition	Volatility	Saving	Ruin
Benchmark	ACO 33/67	25.28%	10.00%	6.98%
Benchmark	ACO 33/67, 175%	43.99%	9.64%	14.81%
Proportional	Four sleeves	26.47%	6.26%	2.25%
	Without global bonds	34.84%	4.56%	2.13%
	Without gold	23.93%	5.72%	2.42%
	Without managed futures	30.23%	7.79%	3.71%
Equal-weight	Four sleeves	25.31%	7.16%	2.79%
	Without global bonds	32.73%	5.63%	2.96%
	Without gold	19.69%	5.73%	2.04%
	Without managed futures	30.70%	10.14%	5.57%

Note: 10,000 paired lifecycle simulations, 1927–2025. Removing a sleeve reallocates its notional pro rata among the other three sleeves, while holding gross exposure at 175%. Saving is the contribution rate that matches ACO utility at a 10% saving rate. ACO holds 100% equities; the common 175% exposure applies to the complete and ablated diversified portfolios. Source: sleeve-ablation dataset.

Every three-sleeve allocation in Table 4 has lower simulated ruin than ACO's 6.98%. Five of the six also match benchmark utility with less than 10% saving. Across the proportional ablations, ruin ranges from 2.13% to 3.71% and equivalent saving from 4.56% to 7.79%. In the equal-weight family, the versions without global bonds or gold also improve both outcomes. The version without managed futures reduces ruin to 5.57%, while requiring 10.14% saving to match benchmark utility. The lifecycle gains therefore extend across several compositions and remain visible when any one of the three additional sleeves is removed.

The comparison with the complete portfolios clarifies the roles of the individual sleeves. Managed futures contribute most to capital preservation: including the proxy lowers ruin by 1.46 and 2.78 percentage points relative to the respective reallocations without it. The paired 95% Monte Carlo intervals for these reductions are [1.21, 1.71] and [2.44, 3.12] points. Global bonds reduce annual volatility relative to both reallocations without them. The equal-weight version without gold improves all three reported metrics relative to its complete portfolio, illustrating that the simple four-sleeve rules leave room for alternative compositions. These comparisons measure each sleeve together with the specified reallocation of its weight; their effects are not additive contributions to the total gain over ACO. All six ablations also have lower ruin and annual volatility than ACO at 175% exposure. Five require less saving than its 9.64%; the equal-weight reallocation without managed futures requires 10.14%.

5.4 Capital preservation without portfolio borrowing

At 100% gross exposure, both diversified families improve capital preservation without a portfolio-level loan. The equal-weight rule has 14.96% annual volatility and 6.21% ruin, against 25.28% and 6.98% for ACO. The proportional rule has 15.59% volatility and 5.22% ruin. These volatilities are about 60% of the benchmark's full-panel value. With $G = 1$, the financing-spread term $\max(G -$

$1, 0)\phi$ is zero, so these results do not depend on that spread. Fees, hedge costs and the derivative exposures inside managed futures remain.

The utility comparison differs. Equivalent saving rises to 14.00% for the equal-weight rule and 12.82% for the proportional rule, against 10% for ACO. Lower ruin therefore does not imply higher expected utility: the latter also values retirement consumption and bequests. In these fixed families, capital preservation improves before borrowing; additional exposure is needed to reach benchmark utility at the same saving rate.

5.5 Historical windows and concentration of variance

We retain 1927–2025 as the primary sample. Large currency movements in the JST observations for Italy 1942 and France 1946 propagate into converted foreign returns. We retain these observations and measure their influence through separate source-event exclusions.

Italy 1942 accounts for 48.35% of ACO's sum of squared deviations from its full-panel mean, and 53.26% for the equal-weight 175% rule. Deleting this common row solely as a diagnostic reduces their volatilities to 18.17% and 17.30%, from 25.28% and 25.31%. Their respective top-five concentration shares are 54.61% and 65.38%; those sets need not contain identical rows. The full-panel volatility comparison is therefore strongly influenced by these observations, whereas the ruin and utility comparisons favor the diversified rule both with and without them.

Table 5 applies the three influence diagnostics consistently. A source event is removed from the resident bootstrap row and from every reconstructed international-equity and global-bond basket; the country diagnostic removes Italy from those baskets in all years and from the resident rows. Under the Italy-1942 event removal, the equal-weight 175% rule has 2.61% ruin and a 7.02% equivalent saving rate, versus 7.19% and 10% for ACO. Its annual volatility is 17.30%, against 18.17% for ACO. Removing France 1946 as well, or Italy in every year, changes the levels but not the direction of the comparison. Across all three diagnostics, the diversified rule combines lower volatility, lower ruin, and a lower saving requirement.

Table 5: Source-event and source-country influence diagnostics

Diagnostic	Portfolio	Volatility	Ruin	Equivalent saving
Italy 1942 event	ACO	18.17%	7.19%	10.00%
	ACO 175%	31.26%	15.79%	10.02%
	Equal-weight 175%	17.30%	2.61%	7.02%
Italy 1942; France 1946 events	ACO	17.50%	7.17%	10.00%
	ACO 175%	29.90%	15.21%	10.28%
	Equal-weight 175%	15.61%	2.74%	7.03%
Italy, all years	ACO	18.08%	7.54%	10.00%
	ACO 175%	31.28%	15.22%	9.67%
	Equal-weight 175%	17.31%	2.70%	7.16%

10,000 paired simulations per diagnostic. The corresponding 95% paired Monte Carlo intervals for the equal-weight ruin difference are $[-5.06, -4.10]$, $[-4.91, -3.95]$, and $[-5.33, -4.35]$ percentage points. This table can be reproduced using the replication commands in the source provenance log.

The two later windows in Table 6 remove entire early regimes. The windows and source-influence diagnostics were selected after examining the full-panel results and serve as historical sensitivities. The fixed 200% rules retain lower ruin and equivalent saving in each window, while their volatility relative to ACO changes.

Table 6: Fixed allocations across historical windows

Window	Portfolio	Volatility	Ruin	Equivalent saving
1927–2025	ACO	25.28%	6.98%	10.00%
1927–2025	ACO 200%	50.28%	20.28%	15.66%
1927–2025	Proportional 200%	30.19%	1.99%	5.09%
1927–2025	Equal-weight 200%	28.86%	2.60%	5.92%
1950–2025	ACO	16.59%	5.42%	10.00%
1950–2025	ACO 200%	32.71%	18.85%	18.09%
1950–2025	Proportional 200%	16.67%	0.43%	5.53%
1950–2025	Equal-weight 200%	15.92%	0.72%	6.93%
1970–2025	ACO	17.24%	6.67%	10.00%
1970–2025	ACO 200%	34.00%	23.67%	29.86%
1970–2025	Proportional 200%	16.78%	0.16%	3.71%
1970–2025	Equal-weight 200%	16.16%	0.21%	3.95%

10,000 paired simulations per window; 1,561, 1,213 and 894 country-years, respectively. Fixed-notional hedges and baseline costs throughout.

5.6 Exposure and implementation sensitivity

Increasing exposure to 200% further reduces ruin and equivalent saving. The proportional rule reaches 1.99% ruin and 5.09% equivalent saving; the equal-weight rule reaches 2.60% and 5.92%. Their full-panel annual volatilities rise to 30.19% and 28.86%, respectively, compared with 25.28% for ACO. Higher exposure thus improves the lifecycle outcomes while increasing annual return dispersion.

Table 7 records the checks most relevant to the construction. A return-dependent screen lowers measured volatility substantially. We report it as a construction diagnostic alongside the unscreened baseline. Reverting to the former bootstrap end treatment or to the idealized hedge changes levels modestly but does not reverse the ranking. A leave-one-source-country-out exercise removes the country from resident observations and from the global equity and bond baskets. Across the 16 omissions, benchmark ruin ranges from 7.10% to 8.66%, while the two 200% sensitivity endpoints remain below it (ranges 1.06–3.18% and 1.72–3.54%). This supports robustness of the ordering, while also showing that its absolute risk depends on the historical sample.

At a 100-basis-point financing spread, the 200% proportional and equal-weight sensitivity rules have ruin of 2.65% and 3.53%, respectively; at 200 basis points this rises to 4.28% and 5.70%. Higher financing costs erode the gains, although both rules retain lower ruin than the 6.98% equity benchmark at these spreads.[2]

[2] Complete numerical outputs, replication scripts, and scope limitations regarding synthetic series are archived in the online repository and detailed in Appendix G.

Table 7: Implementation checks

Specification	ACO ruin	Proportional 200% ruin	Equal-weight 200% ruin
Corrected baseline	6.98%	1.99%	2.60%
Legacy bootstrap end treatment	7.38%	1.98%	2.60%
Ideal full-value hedge	6.98%	1.98%	2.60%
Legacy return screen (diagnostic only)	7.38%	1.68%	2.32%
Leave-one-source-country-out range	7.10–8.66%	1.06–3.18%	1.72–3.54%

Note: First four rows use 10,000 paired simulations; the leave-one-out range uses 5,000 per omission. The 200% columns assess robustness of the portfolio ordering; their volatility exceeds that of ACO in the baseline.

6 Conclusion

The matched-exposure comparison separates diversification from borrowing. Levering ACO to 175% produces 14.81% simulated ruin, versus 2.25% and 2.79% for the diversified families, with higher saving requirements. The advantage of broader asset-class exposure therefore remains when the equity benchmark uses the same amount of portfolio borrowing.

Broadening the investment universe changes the comparison of fixed lifecycle allocations. Our annual public-data implementation recovers the advantage of international over domestic equities. Combining global bonds, gold, and managed futures then improves retirement outcomes: the equal-weight portfolio at 175% exposure reduces simulated ruin from 6.98% to 2.79% and matches benchmark utility with a saving rate of 7.16% rather than 10%.

Diversification and exposure make distinct contributions. At 100% exposure, both diversified families reduce volatility and ruin without portfolio-level borrowing. Additional exposure raises retirement resources and brings the saving requirement below that of the equity benchmark. Under all three source-influence diagnostics, the equal-weight 175% portfolio also has lower annual volatility than ACO, alongside lower ruin and equivalent saving. The later historical windows preserve the favorable lifecycle ranking. All six three-sleeve ablations retain lower ruin than ACO, and five also require less saving to match its utility. The gains thus extend across several diversified compositions, with managed futures making the largest contribution to capital preservation in the ablation comparisons.

The economic implication is that the choice of asset classes and the choice of total exposure belong together in lifecycle portfolio design. In the allocations studied here, international equity diversification provides a strong starting point; diversification across asset classes improves capital preservation, and scaling that diversified portfolio improves the financing of retirement.

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A Asset-Class Return Histories

The cumulative-return exhibit reports annual U.S.-resident returns under the current fixed-notional hedge. The logarithmic axis makes proportional changes comparable across dates.

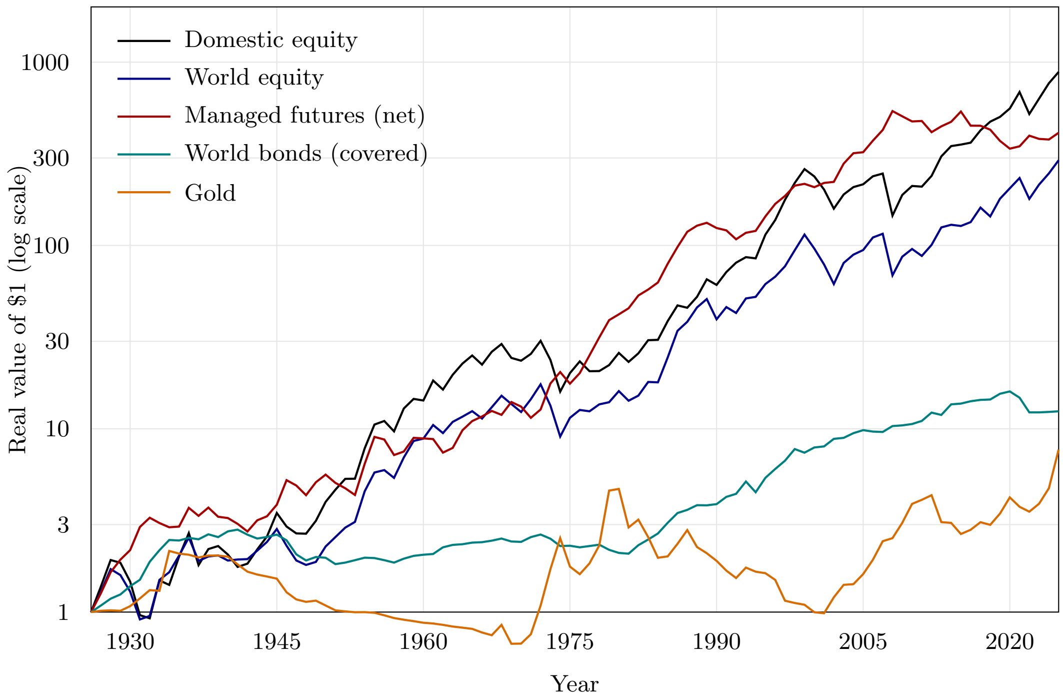

Figure 1: Cumulative real return by asset class, U.S. resident, 1927–2025. Managed futures are the reconstructed time-series-momentum proxy net of the 0.85% fee and 0.57% turnover cost. World bonds are the covered equal-weight basket of all panel sovereigns. Gold is the spliced MeasuringWorth–LBMA price less custody. Equity and gold are unhedged, while bonds and managed futures are currency-hedged under covered interest parity. The managed-futures proxy compounds at close to the equity rate over the century, but its decade-by-decade path is far more volatile before 1960, when the underlying market set is thin.

B Properties of the reconstructed managed-futures proxy

This appendix documents the managed-futures rule and compares its returns with commercial indexes and funds. The rule combines momentum signals at several horizons and scales positions using lagged volatility. The economic motivation follows Moskowitz, Ooi, and Pedersen (2012) and Hurst, Ooi, and Pedersen (2017). The external comparisons measure how closely the proxy captures the return variation of commercial trend-following strategies over their common history.

B.1 Monthly inputs and the annual simulation series

The annual lifecycle panel and the monthly signal inputs are distinct data layers. The monthly snapshot supplies the managed-futures construction; complete calendar years are then compounded for the annual simulation. The source table records the original reconstruction's inputs and transformations. Price-only equities, synthetic bond returns, omitted commodity carry and roll, and incomplete early currency coverage limit the economic interpretation of the reconstructed return.

The annual currency treatment applies a forward to the initial foreign principal. Exchange-rate exposure on the subsequent asset gain or loss remains. This differs from removing foreign collateral and replacing it with the resident bill for the entire terminal asset value.

Table 8: Monthly data used to construct the managed-futures proxy

Input	Monthly source and transformation	What enters the MF rule
Equities	National price indexes from OECD Main Economic Indicators via FRED, and before their availability, the U.S., U.K., German, and French NBER Macrohistory monthly price series.	18 local-currency price returns. Dividends are absent.
Rates	Monthly sovereign yields from OECD via FRED, augmented before their start by NBER U.S. yields, U.K. Consol/Bank of England yields, and the French 3% rente. Each yield pair is priced as a par ten-year bond, then the lagged three-month cash return is subtracted.	18 synthetic local-currency bond excess returns, not changes in yields and not observed futures returns.
Commodities	NBER/BLS monthly wholesale prices for the early history and the World Bank Pink Sheet from 1960. Gold uses the official U.S. price before 1960 and the Pink Sheet thereafter, and silver starts in 1960.	19 commodity spot-price returns plus gold and silver. Carry, basis, and futures roll are unavailable and therefore set to zero.
Currencies	Board of Governors H.10 daily dollar quotations via FRED, reduced to the last available observation in each month, with a lagged three-month interest differential forming a synthetic forward excess return.	AUD, CAD, CHF, GBP, JPY from 1971 and EUR from 1999. Earlier European currencies are not backfilled.
Collateral and deflator	Monthly three-month cash rates from OECD via FRED, with a lagged prior-year JST/GMD fallback only where monthly rates do not exist, plus monthly U.S. CPI for deflation.	U.S. cash collateral is added once to the USD MF NAV, and CPI converts the final monthly USD return to real terms.

Note: The machine-readable source-segment file records the provider, series identifier, dates, transformation, and source seam for every market. The monthly snapshot is frozen for this replication. The first month of each source segment receives no return, preventing differences in source price levels from entering the trading signal.

B.2 Comparison with commercial trend-following indexes

These correlations compare monthly dollar returns independently of the resident panel, annual hedge, and lifecycle bootstrap. The external comparison places the net monthly proxy alongside SG CTA, SG Trend, Barclay BTOP50, and the KMLMSIM simulation over their common period. The indexes are external benchmarks and enter no lifecycle resampling. Their proprietary inputs remain subject to their source terms.

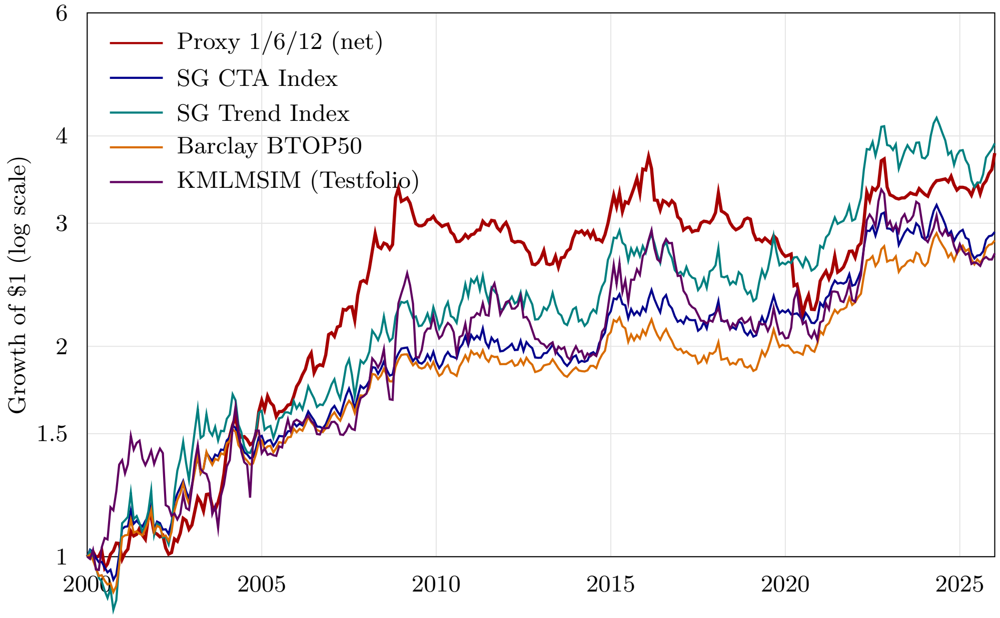

Figure 2: Growth of one dollar, monthly, 2000–2025: the net managed-futures proxy (variant 1/6/12, the one used in the paper) against three commercial trend-following indexes and KMLMSIM, the Testfolio simulation of KMLM. SG CTA, SG Trend, and BTOP50 are net of index fees, and the proxy is net of the 0.85% management fee and 0.57% turnover cost. KMLMSIM uses the KFA MLM Index less 0.90% per year before the ETF's inception. The SG and BarclayHedge indexes are proprietary and are used here only under the terms held for this research.

Table 9: Monthly correlation of the net proxy with trend-following benchmarks

Benchmark	Corr.	Common window
SG CTA Index	0.51	2000-01–2025-12 (312 mo.)
SG Trend Index	0.49	2000-01–2025-12 (312 mo.)
Barclay BTOP50 Index	0.53	2000-01–2025-12 (312 mo.)
KMLMSIM (Testfolio simulation)	0.52	2000-01–2025-12 (312 mo.)
DBMF ETF (live)	0.61	2019-06–2025-12 (79 mo.)
KMLM ETF (live)	0.63	2021-01–2025-12 (60 mo.)

Memo, 2000–2025: the three commercial indexes correlate 0.96–0.97 with each other, whereas KMLMSIM correlates 0.60 with SG CTA. The proxy near 0.51 sits between these two reference patterns.

Note: contemporaneous Pearson correlation of monthly returns over the common months shown. KMLMSIM is Testfolio's KFA MLM Index-based simulation of KMLM before the ETF's inception, and the "live" rows use the ETFs' own return history from the first full month after listing.

B.3 The proxy among commercial benchmarks and investable funds

The broader comparison retains the original pairwise correlation matrix, the matrix on a common window, and the regressions on the proxy. Pairwise windows differ when funds have different inception dates. The common-window matrix addresses that comparability issue, at the cost of a shorter history. The regressions report exposures to the proxy, intercepts, and tracking errors, describing both common return variation and differences across products.

Table 10: Correlation matrix of monthly returns: the net proxy, commercial benchmarks, and investable managed-futures funds

	SG CTA	SG Trend	BTOP50	KMLMSIM	DBMF	KMLM	QMHIX	AHLIX	AHLT	IMF	ISMF	WTMF	APEX
Proxy	**0.51**	**0.49**	**0.53**	**0.52**	**0.61**	**0.63**	**0.52**	**0.54**	**0.65**	**0.63**	**0.58**	**0.23**	**0.15**
SG CTA	—	0.97	0.97	0.60	0.86	0.77	0.83	0.84	0.86	0.76	0.87	0.40	0.50
SG Trend	0.97	—	0.96	0.60	0.86	0.74	0.85	0.86	0.85	0.72	0.89	0.42	0.46
BTOP50	0.97	0.96	—	0.58	0.85	0.73	0.79	0.83	0.84	0.69	0.82	0.41	0.57
KMLMSIM	0.60	0.60	0.58	—	0.66	1.00	0.75	0.63	0.69	0.89	0.86	0.34	0.33
DBMF	0.86	0.86	0.85	0.66	—	0.68	0.73	0.74	0.77	0.29	0.53	0.24	0.50
KMLM	0.77	0.74	0.73	1.00	0.68	—	0.74	0.65	0.69	0.89	0.86	0.12	0.33
QMHIX	0.83	0.85	0.79	0.75	0.73	0.74	—	0.70	0.74	0.54	0.82	0.33	0.40
AHLIX	0.84	0.86	0.83	0.63	0.74	0.65	0.70	—	0.85	0.76	0.83	0.39	0.28
AHLT	0.86	0.85	0.84	0.69	0.77	0.69	0.74	0.85	—	0.82	0.90	0.13	0.56
IMF	0.76	0.72	0.69	0.89	0.29	0.89	0.54	0.76	0.82	—	0.86	0.28	0.40
ISMF	0.87	0.89	0.82	0.86	0.53	0.86	0.82	0.83	0.90	0.86	—	0.42	0.59
WTMF	0.40	0.42	0.41	0.34	0.24	0.12	0.33	0.39	0.13	0.28	0.42	—	0.01
APEX	0.50	0.46	0.57	0.33	0.50	0.33	0.40	0.28	0.56	0.40	0.59	0.01	—

Note: contemporaneous Pearson correlation of monthly returns, each pair over the overlap of its two histories within 2000-01 to 2025-12, the frame of the appendix. The first month in that frame is 2000-01 for the SG indexes (their full history), 2019-06 for DBMF, 2021-01 for KMLM, 2011-02 for WTMF, 2013-08 for QMHIX, 2014-09 for AHLIX, 2022-04 for Apex, 2023-10 for AHLT, and 2025-04 for IMF and ISMF; BTOP50 (from 1987) and KMLMSIM (from 1988) are restricted to the common frame, and the proxy is available from 1926-03, so the number of overlapping months of a pair is determined by the later of its two start months. The index returns are the proprietary SG and BarclayHedge series and the testfol.io series of Table 9; fund returns are total returns net of fund fees from Yahoo Finance adjusted closes. Shading is proportional to the correlation; the horizontal rule separates the benchmark block from the investable funds. QMHIX, AHLIX, AHLT, IMF, ISMF, DBMF, and KMLM are pure trend-following products; WTMF is a systematic equity-timing strategy and Apex a multi-strategy fund.

Table 11: Correlation matrix over a common window: the proxy and the benchmarks and funds with at least ten years of history

	SG CTA	SG Trend	BTOP50	KMLMSIM	WTMF	QMHIX	AHLIX
Proxy	**0.57**	**0.57**	**0.60**	**0.56**	**0.23**	**0.52**	**0.54**
SG CTA	—	0.98	0.97	0.72	0.39	0.84	0.84
SG Trend	0.98	—	0.96	0.72	0.40	0.86	0.86
BTOP50	0.97	0.96	—	0.69	0.41	0.79	0.83
KMLMSIM	0.72	0.72	0.69	—	0.31	0.75	0.63
WTMF	0.39	0.40	0.41	0.31	—	0.33	0.39
QMHIX	0.84	0.86	0.79	0.75	0.33	—	0.70
AHLIX	0.84	0.86	0.83	0.63	0.39	0.70	—

Note: contemporaneous Pearson correlation of monthly returns, every pair evaluated over the same window 2014-09 to 2025-12 (136 months), the maximal common window of the eight series (proxy from 1926-03; SG indexes from 2000-01, BTOP50 and KMLMSIM restricted to the common frame; WTMF from 2011-02, QMHIX from 2013-08, AHLIX from 2014-09). DBMF (79 months) and KMLM (60 months) do not meet the ten-year rule. Sources and conventions as in Table 10.

Table 12: Regressions of managed-futures benchmarks and investable funds on the net proxy

Series	n	α (%/mo.)	β	R^2	σ_f/σ_p	TE (%/yr.)
SG CTA	312	+0.18 (+1.4)	0.41 (+10.6)	0.26	0.80	9.8
SG Trend	312	+0.22 (+1.2)	0.60 (+9.9)	0.24	1.24	12.4
BTOP50	312	+0.18 (+1.6)	0.39 (+10.9)	0.28	0.74	9.5
KMLMSIM	312	+0.06 (+0.3)	0.76 (+10.9)	0.28	1.44	13.5
DBMF	79	+0.45 (+1.6)	0.62 (+6.8)	0.38	1.00	9.8
KMLM	60	−0.13 (−0.3)	0.78 (+6.1)	0.39	1.24	10.2
QMHIX	149	+0.30 (+0.9)	0.73 (+7.3)	0.27	1.42	13.4
AHLIX	136	+0.28 (+1.4)	0.47 (+7.4)	0.29	0.87	10.1
AHLT	27	−0.53 (−0.8)	1.49 (+4.3)	0.42	2.30	12.0
IMF	9	−2.08 (−1.5)	1.10 (+2.2)	0.40	1.73	11.9
ISMF	9	+0.57 (+0.9)	0.43 (+1.9)	0.34	0.74	7.3
WTMF	179	+0.04 (+0.3)	0.14 (+3.1)	0.05	0.62	10.7
APEX	45	+1.11 (+2.9)	0.13 (+1.0)	0.02	0.87	12.5

Note: ordinary least squares $R^s_t = \alpha + \beta R^{proxy}_t + \varepsilon_t$ on monthly returns, each series over its overlap with the proxy within 2000-01 to 2025-12 (n column; all windows end in 2025-12). α is in per cent per month; α and β carry Student t statistics in parentheses. σ_f/σ_p is the ratio of annualised volatilities, and TE the tracking error: the annualised standard deviation of the active return $R^s_t - R^{proxy}_t$, in per cent per year. Sources and conventions as in Table 10.

B.4 Annual properties against the other asset classes

The annual moments and scatter plots use the current U.S.-resident panel and fixed-notional hedge. The proxy is net of a 0.85% management fee and a 0.57% annual turnover deduction. The exhibits describe the proxy's relation to the other asset classes, including its behavior in equity-loss years.

Table 13: The managed-futures proxy against the other asset classes

Asset	Mean	SD	Sharpe_0	Skew	5th pctl	Max DD
Domestic equity	8.81%	18.69%	0.47	-0.24	-21.4%	-51.9%
World equity	7.42%	17.48%	0.42	-0.25	-23.1%	-48.1%
World bonds (covered)	2.82%	7.03%	0.40	+0.15	-7.6%	-35.2%
Gold	3.70%	19.70%	0.19	+1.88	-21.2%	-79.0%
Managed-futures proxy (net)	7.17%	14.05%	0.51	+0.44	-13.1%	-37.6%

Year	World equity	MF proxy	Gold	U.S. inflation
1930	-18.7%	12.7%	6.0%	-2.3%
1931	-29.5%	33.8%	10.2%	-9.0%
1937	-25.4%	-9.6%	-3.4%	3.6%
1946	-18.9%	36.3%	-15.7%	8.3%
1973	-23.5%	38.6%	58.1%	6.3%
1974	-32.2%	15.4%	47.6%	11.0%
1990	-22.5%	-6.3%	-9.2%	4.9%
2002	-21.8%	1.4%	22.0%	1.6%
2008	-40.7%	26.7%	3.9%	3.8%
2022	-23.1%	14.4%	-6.0%	8.0%

Note: U.S. resident, annual real returns, 1927–2025. *Top panel:* $\text{Sharpe}_{r=0}$ is the mean over the standard deviation with a zero real cash rate, and "Max DD" is the largest peak-to-trough decline in cumulative real value. *Bottom panel:* the ten worst years for world equity and the proxy's realized return in each, where "U.S. inflation" is the same year's CPI change.

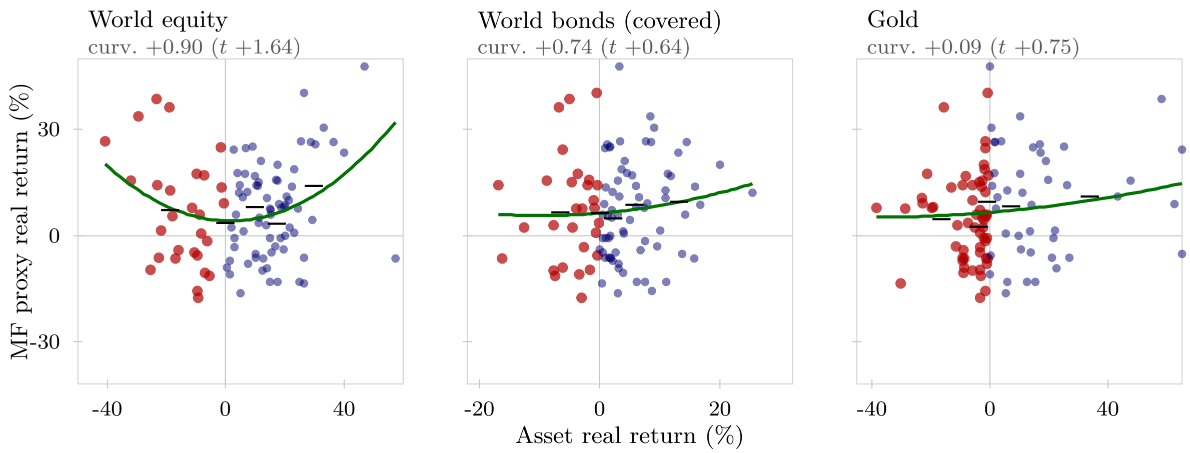

Figure 3: Annual real return of the net managed-futures proxy against the annual real return of each asset class, U.S. resident, 1927–2025. Each point is one year, with red points the years the asset itself lost and blue points the years it gained. The green curve is a descriptive quadratic fit and the black ticks are the mean proxy return within each quintile of the asset's return. "curv." is the coefficient on the squared term, and t is its heteroskedasticity-robust (White HC1) statistic.

B.5 Sensitivity to the construction choices

The series-level comparison varies signal horizons and management fees, using each variant's measured turnover cost. It describes the underlying proxy series independently of the lifecycle bootstrap and resident-currency hedge.

Table 14: Managed-futures proxy under alternative construction choices

Variant	Mean	SD	Sharpe_0	Skew	5th pctl	Corr. baseline
1/6/12 blend (baseline)†	7.04%	13.93%	0.51	+0.39	-12.1%	1.000
1/3/12 blend	7.25%	14.33%	0.51	+0.60	-13.1%	0.961
12-month signal only	6.96%	13.76%	0.51	+0.04	-17.4%	0.831
1-month signal only	5.21%	12.74%	0.41	+0.66	-12.5%	0.772
1/6/12 blend, fee doubled to 1.70%	6.13%	13.81%	0.44	+0.39	-12.9%	1.000

Note: U.S. resident, annual real returns, 1927–2025, net of each variant's management fee and its own realized turnover cost. "Corr. baseline" is the Pearson correlation of the annual series with the 1/6/12 blend used throughout the paper. The baseline mean here differs from Table 13 by about 0.15 point because that table applies a flat annual turnover drag while this one compounds the monthly realized cost. The correlations are unaffected by the cost convention. †Baseline.

C Leverage ladders

The following tables use the 1,561-country-year panel, the ACO-style block continuation, and the fixed-notional forward convention. Each family has five specified gross-exposure levels. Volatility is measured after the weights are set; the tables do not estimate a volatility target. They use 10,000 paths at seed 20260827 and the same paired ACO benchmark. The benchmark has 25.28% annual volatility, 6.98% retirement ruin, and a 10% reference saving rate. Source-influence diagnostics in Table 5 examine the observations that contribute most to pooled variance.

Table 15: Proportional leverage ladder

Gross	Equity/Bonds/Gold/MF	Volatility	Saving	Ruin
100%	40/26.67/16.67/16.67	15.59%	12.82%	5.22%
125%	50/33.33/20.83/20.83	19.15%	9.90%	3.39%
150%	60/40/25/25	22.78%	7.81%	2.66%
175%	70/46.67/29.17/29.17	26.47%	6.26%	2.25%
200%	80/53.33/33.33/33.33	30.19%	5.09%	1.99%

Note: Equity is the ACO 33/67 sleeve. Bonds and managed futures use the fixed-notional hedge. Saving is the contribution rate that matches benchmark utility. Tabulated from the leverage ladder simulation dataset.

Table 16: Equal-weight leverage ladder

Gross	Equity/Bonds/Gold/MF	Volatility	Saving	Ruin
100%	25/25/25/25	14.96%	14.00%	6.21%
125%	31.25/31.25/31.25/31.25	18.33%	10.95%	4.20%
150%	37.5/37.5/37.5/37.5	21.80%	8.78%	3.27%
175%	43.75/43.75/43.75/43.75	25.31%	7.16%	2.79%
200%	50/50/50/50	28.86%	5.92%	2.60%

Note: Equity is the ACO 33/67 sleeve. Bonds and managed futures use the fixed-notional hedge. Saving is the contribution rate that matches benchmark utility. Tabulated from the leverage ladder simulation dataset.

D Historical-sample uncertainty

Monte Carlo intervals measure sampling noise conditional on the observed history and the specified model. They do not measure uncertainty about the historical return-generating process. We retain 1927–2025 as the main sample. The 1950–2025 and 1970–2025 experiments, each with 10,000 paths, address sensitivity to the historical window. They are reported in the main text and archived in the sample window directory.

The source-country omissions remove the country from resident rows, international-equity constituents, and world-bond constituents. At 5,000 paths per omission, ACO ruin ranges from 7.10% to 8.66%; the proportional 200% family ranges from 1.06% to 3.18%, and the equal-weight 200% family from 1.72% to 3.54%. The corresponding saving rates range from 5.00% to 5.26% and from 5.63% to 6.16%, against each sample's 10% reference. These are ranges across omissions, not confidence intervals, documented in the country sensitivity audit.

The separate event experiments remove Italy 1942, then Italy 1942 and France 1946; the country experiment removes Italy throughout. Their inputs and outputs are archived in the source exclusion directory. They measure the influence of observations identified during the audit. The exclusions apply to resident rows and global equity and bond constituents; the monthly managed-futures inputs are held fixed. The main text reports these results.

E Risk-aversion sensitivity

The exercise holds portfolio weights and exposure fixed and recomputes the unlevered ACO utility target at every risk-aversion coefficient. We report both the earlier joint calibration, which varies the annual bequest coefficient with γ, and a control holding that coefficient at 2360. The latter isolates the effect of changing curvature without changing the annual bequest coefficient. Neither exercise reoptimizes portfolio weights. Ruin is a wealth-depletion event and is unchanged by preferences alone.

Table 17: Risk aversion and the saving requirement at 200% exposure

Calibration	γ	ACO	Proportional	Equal-weight
Joint calibration	2.0	10.35%	2.38%	3.23%
Joint calibration	3.0	20.09%	3.73%	4.59%
Joint calibration	3.84	15.66%	5.09%	5.92%
Joint calibration	5.0	20.25%	4.90%	5.73%
Joint calibration	7.5	20.26%	4.64%	5.48%
Joint calibration	10.0	17.60%	4.53%	5.38%
Fixed annual bequest	2.0	9.23%	2.96%	3.86%
Fixed annual bequest	3.0	12.42%	4.97%	5.80%
Fixed annual bequest	3.84	15.66%	5.09%	5.92%
Fixed annual bequest	5.0	20.26%	4.90%	5.73%
Fixed annual bequest	7.5	20.26%	4.64%	5.48%
Fixed annual bequest	10.0	17.60%	4.53%	5.38%

10,000 paired paths; ACO at 100% requires 10% saving at each γ. The joint calibration uses $\theta = 2360\,12^{3.84-\gamma}$; the fixed-annual calibration keeps $\theta = 2360$. At $\gamma = 3.84$ both coincide. All portfolios keep their weights and exposure fixed. Ruin is 20.28%, 1.99%, and 2.60%, respectively, throughout both grids.

The modest saving response for diversified portfolios survives the fixed-coefficient control. At $\gamma = 10$, consumption accounts for virtually all of their negative expected utility: the shifted bequest term contributes less than 10^{-10} of its magnitude. For the equal-weight rule, the worst one percent of paths accounts for 90.6% of negative utility, compared with 4.8% at $\gamma = 2$. High curvature therefore does give substantial weight to adverse consumption outcomes. Social Security and the consumption floor, the distinction between upside return variation and consumption shortfalls, and the changing benchmark utility explain why this need not translate into an exploding saving requirement. The relative utility residual at the reported saving solutions is below 10^{-6} throughout the fixed-coefficient grid. Tail concentration at high γ also makes those estimates more sensitive to the sampled adverse paths.

F Contribution- and withdrawal-rate sensitivity

The policy exercise varies contribution rates at a fixed 4% withdrawal rule and withdrawal rates at a fixed 10% benchmark contribution. On the contribution axis, the comparison is anchored to benchmark utility at that row's contribution rate. On the withdrawal axis, it is anchored to the benchmark under that row's withdrawal rule. Thus saving rates should be compared within a row.

With proportional contributions and an initial-wealth-based withdrawal rule, changing the contribution rate scales account wealth and withdrawals together. This scaling leaves ruin unchanged while increasing consumption and bequests. Higher withdrawal rates increase depletion in all three portfolios. Both diversified families at 200% exposure retain lower simulated ruin and equivalent saving over this grid.

Table 18: Contribution- and withdrawal-rate sensitivity of the fixed diversified families

Rate	ACO 33/67		Proportional 200%		Equal-weight 200%	
	Ruin	Saving	Ruin	Equiv. saving	Ruin	Equiv. saving
lower is better, the ACO columns are the reference						
Contribution rate r_c (withdrawal held at 4%)						
$r_c = 5\%$	*6.98%*	*5.00%*	1.99%	2.67%	2.60%	3.07%
$r_c = 10\%^\dagger$	*6.98%*	*10.00%*	1.99%	5.09%	2.60%	5.92%
$r_c = 15\%$	*6.98%*	*15.00%*	1.99%	7.36%	2.60%	8.62%
Withdrawal rate r_w (contribution held at 10%)						
$r_w = 3\%$	*2.54%*	*10.00%*	0.95%	5.48%	1.16%	6.32%
$r_w = 4\%^\dagger$	*6.98%*	*10.00%*	1.99%	5.09%	2.60%	5.92%
$r_w = 5\%$	*15.47%*	*10.00%*	4.81%	4.34%	6.40%	5.12%

Note: 10,000 paired lifecycle paths per row, full 1,561-country-year panel, ten-year mean stationary blocks, $\phi = 30$ bp and $\kappa = 10$ bp. Entries are levels, in percent: retirement ruin and the utility-equivalent saving rate, both lower is better. The two ACO 33/67 columns are the reference. The two 200% families keep their fixed weights and gross exposure in every row. On the contribution-rate axis the ACO 33/67 benchmark also contributes r_c, so its reference saving rate is r_c itself and the family saving is the rate that matches ACO's lifecycle utility at that r_c. On the withdrawal-rate axis the benchmark saving rate stays at 10%. †Baseline.

G Reproducibility and Conversion Checks

The baseline is identified by 1,561 resident country-years, 10,000 paths, seed 20260827, ten-year mean blocks, and the fixed-notional hedge convention. Portfolios share lifetime draws within each historical panel; changing the panel changes the draws even at the same seed. Full-panel numerical outputs are archived in the core results and leverage ladder datasets. The constant-exposure sleeve ablations are stored in the ablation dataset; the replication script records the reallocation rule and all simulation settings. Historical-window outputs and replication commands are recorded in the sample window directory. The variance concentration script checks baseline volatility agreement before reporting concentration and deletion diagnostics.

The annual conversion and simulation can be reproduced from the supplied annual inputs. The monthly managed-futures reconstruction has an upstream engine dependency outside this paper directory. The package consumes the frozen monthly output of that engine. Reconstructing it from provider data requires the upstream engine, while regenerating the external-index comparisons requires the licensed inputs.

For a foreign asset's local real return R_f^L and inflation π_f, the nominal gross return is $(1 + R_f^L)(1+\pi_f)$. A fixed-notional forward covers the initial foreign principal. If q is the realized resident-currency value of one unit of foreign currency relative to its initial value, G_f the foreign nominal gross asset return, and B_c/B_f the covered-interest-parity forward factor, the hedged nominal gross payoff is

$$G^H_{f\to c} = \frac{B_c}{B_f} + (G_f - 1)q.$$

Deflating by resident inflation produces the real return used in the panel. The second term retains currency exposure on the foreign gain or loss. Consequently, equality of all residents' asset excess returns is not a valid check for this convention. That identity belongs to the earlier idealized full-terminal-value hedge.

The return-cap diagnostic, ideal hedge, legacy block restart, restricted historical windows, and source exclusions are distinct experiments. Their output files identify the changed convention. The baseline retains large returns. Documented market-year exclusions and source-influence diagnostics are recorded separately.

H Margin calls on the do-it-yourself route

Direct portfolio borrowing introduces a maintenance-margin constraint. We evaluate six directly financed allocations using the 1,561-country-year panel, fixed-notional currency hedges, and the baseline fees and costs. The account rebalances to its target exposure at the start of each year, and its equity share is tested against a 25% maintenance threshold at year-end, before the next rebalance. We draw 10,000 paired 86-year return paths with ten-year mean blocks, ACO-style continuation, and seed 20260827. This account diagnostic follows each return path for the full horizon, without household contributions, withdrawals, or mortality truncation.

For initial equity normalized to one, gross assets L, debt $L-1$, asset-book nominal gross return G_A, and debt gross accrual G_D, the year-end equity fraction is

$$m_{\text{end}} = \frac{LG_A - (L-1)G_D}{LG_A}.$$

A maintenance call at threshold m occurs when $G_A < (L-1)G_D/[L(1-m)]$, provided the asset value remains positive. The threshold therefore depends on financing as well as exposure. Nominal rather than real account values determine the margin test. To match the paper's real financing deduction exactly, debt accrues at $G_D = (1 + R_f^{(c)} + \phi)(1 + \pi_c)$. The asset book includes sleeve fees and hedge costs, while the borrowing charge enters the debt leg.

Table 19: Year-end maintenance-margin diagnostic

Allocation	Gross exposure	Threshold breaches	Minimum equity share
ACO 33/67	175%	1,081	21.41%
ACO 33/67	200%	17,934	8.31%
Equal-weight four-sleeve	175%	0	47.69%
Proportional four-sleeve	200%	0	37.59%
Equal-weight four-sleeve	200%	0	38.98%
90/60 covered global bonds	150%	0	53.24%

Note: 860,000 simulated account-years per allocation; maintenance threshold 25%. Minimum equity share is the smallest year-end ratio of account equity to gross assets. Account-years reuse the historical panel through block resampling. This table can be reproduced from the margin simulation dataset using the margin call experiment script.

The four diversified allocations have no year-end threshold breaches. ACO records 1,081 at 175% exposure and 17,934 at 200%, or 0.13% and 2.09% of account-years. Exposure is reset annually in this diagnostic, including after a breach; it measures threshold crossings rather than a lifecycle policy with forced liquidation. Annual rebalancing restores the initial equity cushion: 50% of assets at 200% gross exposure and 57.14% at 175%.

Actual margin management also depends on intrayear price paths, variation margin, liquidity, and changing broker requirements. A fund implementation manages these obligations within the fund. Extending the lifecycle model to account for such funding constraints would connect the portfolio-return comparison to instrument-level implementation.

I Matched-leverage comparisons

ACO retains its 33/67 domestic–international equity mix at each exposure. All borrowed amounts use the resident bill and the same financing spread as the diversified rules. These comparisons hold borrowing constant; portfolio volatility remains an outcome of asset composition.

Table 20: Fixed leverage ladder for ACO 33/67

Gross	Volatility	Equiv. saving	Ruin
100%	25.28%	10.00%	6.98%
125%	31.47%	8.56%	8.13%
150%	37.71%	8.30%	10.69%
175%	43.99%	9.64%	14.81%
200%	50.28%	15.66%	20.28%

10,000 paired paths, 1927–2025. Borrowing costs the resident bill plus the same 30 bp real deduction used for the diversified portfolios. The utility target remains ACO at 100% exposure and 10% saving.

Table 21: Levered ACO in the household-policy sensitivities

Saving	Withdrawal	ACO ruin	ACO equiv.	Prop. equiv.	Equal equiv.
5.00%	4.00%	20.28%	4.63%	2.67%	3.07%
10.00%	4.00%	20.28%	15.66%	5.09%	5.92%
15.00%	4.00%	20.28%	46.06%	7.36%	8.62%
10.00%	3.00%	13.73%	9.84%	5.48%	6.32%
10.00%	4.00%	20.28%	15.66%	5.09%	5.92%
10.00%	5.00%	26.75%	24.11%	4.34%	5.12%

All three portfolios have 200% exposure. The reference is unlevered ACO at the contribution and withdrawal rates in each row. The diversified ruin rates are reported in Table 18.

Table 22: Financing and hedge-cost sensitivities at matched exposure

Spread	Hedge cost	ACO ruin	ACO equiv.	Prop. ruin	Equal ruin
0.30%	0.10%	20.28%	15.66%	1.99%	2.60%
1.00%	0.10%	22.73%	22.03%	2.65%	3.53%
2.00%	0.10%	26.38%	38.35%	4.28%	5.70%
3.00%	0.10%	29.73%	76.63%	6.84%	9.27%
0.30%	0.10%	20.28%	15.66%	1.99%	2.60%
0.30%	0.50%	20.28%	15.66%	2.24%	3.04%
0.30%	1.00%	20.28%	15.66%	2.77%	3.85%

All three portfolios have 200% exposure. ACO equities are unhedged, so hedge-cost changes affect only the diversified sleeves. Equivalent saving targets unlevered ACO at 10% saving.

Table 23: Construction checks including levered ACO

Specification	ACO 100%	ACO 200%	Prop. 200%	Equal 200%
Baseline	6.98%	20.28%	1.99%	2.60%
Block restart	7.38%	19.68%	1.98%	2.60%
Full-value hedge	6.98%	20.28%	1.98%	2.60%
Return screen	7.38%	21.71%	1.68%	2.32%
War screen	6.04%	25.67%	0.43%	0.43%
Source-country range	7.10–8.66%	19.92–24.00%	1.06–3.18%	1.72–3.54%

Retirement ruin; 10,000 paired paths for construction checks and 5,000 per source-country omission. All other assumptions follow the corresponding diagnostic.